

* Practical - 15 *

PU VAULT



* MainActivity.java

```
package com.example.usercrudapp;  
  
public class MainActivity extends AppCompatActivity {
```

```
    EditText etId, etName, etAddress, etContact;  
    Button btnAdd, btnUpdate, btnDelete, btnList;  
    ListView listViewUsers;  
    DBHelper dbHelper;  
    ArrayAdapter<String> adapter;  
    ArrayList<String> userList;  
    ArrayList<Integer> userIdList;
```

④ Override

```
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.activity_main);
```

```
        etId = findViewById(R.id.etId);  
        etName = findViewById(R.id.etName);  
        etAddress = findViewById(R.id.etAddress);  
        etContact = findViewById(R.id.etContact);  
        btnAdd = findViewById(R.id.btnAdd);  
        btnUpdate = findViewById(R.id.btnUpdate);  
        btnDelete = findViewById(R.id.btnDelete);  
        btnList = findViewById(R.id.btnList);  
        listViewUsers = findViewById(R.id.listViewUsers);
```

```
        dbHelper = new DBHelper(this);  
        userList = new ArrayList<>();  
        userIdList = new ArrayList<>();
```



```
btnAdd.setOnClickListener (v -> {  
    if (etName.getText().toString().isEmpty() ||  
        etAddress.getText().toString().isEmpty() ||  
        etContact.getText().toString().isEmpty()) {  
        Toast.makeText(this, "Please fill all fields",  
            Toast.LENGTH_SHORT).show();  
        return;  
    }  
})
```

```
        boolean inserted = dbHelper.insertUser(  
            etName.getText().toString(),  
            etAddress.getText().toString(),  
            etContact.getText().toString());  
    };  
    if (inserted) {  
        Toast.makeText(this, "User Added",  
            Toast.LENGTH_SHORT).show();  
        clearFields();  
        loadUsers();  
    } else {  
        Toast.makeText(this, "Error Adding User",  
            Toast.LENGTH_SHORT).show();  
    }  
});
```

```
btnUpdate.setOnClickListener (v -> {  
    if (etId.getText().toString().isEmpty()) {  
        Toast.makeText(this, "Select a user to update",  
            Toast.LENGTH_SHORT).show();  
        return;  
    }  
})
```



```

        boolean updated = dbHelper.updateUser(
            etId.getText().toString(),
            etName.getText().toString(),
            etAddress.getText().toString(),
            etContact.getText().toString()
        );
        if (updated) {
            Toast.makeText(this, "User Updated",
                Toast.LENGTH_SHORT).show();
            clearFields();
            loadUsers();
        } else {
            Toast.makeText(this, "Error Updating User",
                Toast.LENGTH_SHORT).show();
        }
    });

```

```

btnDelete.setOnClickListener(v -> {
    if (etId.getText().toString().isEmpty()) {
        Toast.makeText(this, "Select a user to delete",
            Toast.LENGTH_SHORT).show();
        return;
    }

```

```

        boolean deleted = dbHelper.deleteUser(etId.getText().
            toString());
        if (deleted) {
            Toast.makeText(this, "User Deleted", Toast.
                LENGTH_SHORT).show();
            clearFields();
            loadUsers();
        } else {

```



```
Toast.makeText(this, "Error Deleting User",  
                Toast.LENGTH_SHORT).show();  
    }  
});  
  
btnList.setOnClickListener(v -> loadUsers());  
  
listViewUsers.setOnItemClickListener((parent, view,  
                                     position, id) -> {  
    int userId = userIdList.get(position);  
  
    Cursor cursor = dbHelper.getUserById(userId);  
    if (cursor != null && cursor.moveToFirst()) {  
        etId.setText(String.valueOf(cursor.getInt(0)));  
        etName.setText(cursor.getString(1));  
        etAddress.setText(cursor.getString(2));  
        etContact.setText(cursor.getString(3));  
        cursor.close();  
    }  
});  
}  
  
private void loadUsers() {  
    userList.clear();  
    userIdList.clear();  
    Cursor cursor = dbHelper.getAllUsers();  
    if (cursor != null) {  
        while (cursor.moveToNext()) {  
            int id = cursor.getInt(0);  
            String name = cursor.getString(1);  
            String address = cursor.getString(2);  
            String contact = cursor.getString(3);
```

```
        userList.add(Cid + " - " + name + " (" +  
                    + contact + ")");  
        userIdList.add(Cid);  
    }  
    cursor.close();  
}  
adapter = new ArrayAdapter<>(this, android.R.layout.  
                             simple_list_item_1,  
                             userList);  
listViewUsers.setAdapter(adapter);  
}  
  
private void clearFields() {  
    etId.setText("");  
    etName.setText("");  
    etAddress.setText("");  
    etContact.setText("");  
}  
}
```



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** Output **

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